

Ayman Kamel Korany Bayoumi

Game Developer (Unreal Engine 5, C++) · Simulation & Systems Engineer

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PROFILE

Mechanical engineer and computer scientist with 8+ years of software development, now focused on game development in **Unreal Engine 5 with C++**. My background in numerical solvers, physics and verification testing carries directly into game systems: abilities, AI, procedural generation, deterministic simulation and networking. I build data-driven, well-tested code with design documents and a clean, honest Git history.

GAME PROJECTS (UNREAL ENGINE 5.8, C++)

DUATFALL · Action roguelite set in the Egyptian underworld · github.com/aymank2020/duatfall

Pre-production: design document and roadmap complete; engine milestones in progress

- Abilities, attributes and boons designed on the Gameplay Ability System; enemy AI in StateTree with a three-phase boss.
- Seeded run generation for reproducible runs; versioned save data with migration tests; 60 fps target on mid-range laptop hardware.

GRIDLINE · Squad turn-based tactics · github.com/aymank2020/gridline

Pre-production: design document and roadmap complete

- Pure C++ rules engine separated from presentation; command pattern with validate/apply; A*, Dijkstra ranges and supercover line of sight.
- Deterministic replay by state hash; Utility AI; planned 100+ automation tests including property tests on random maps.

HEARTHWAKE · Co-op survival crafting (1–4 players) · github.com/aymank2020/hearthwake

Pre-production: design document and roadmap complete

- Server-authoritative replication; FastArray inventory with idempotent operations; structural building stability; seeded PCG world.

EXPERIENCE

Numerical Solver / Structures Task Contributor · Prometheus · 2026

- Developed structural-engineering benchmark tasks for AI coding agents, with reference solutions, tests, scoring logic and reproducible evaluation workflows.
- Worked with 2D frame analysis, finite element methods, P-Delta / geometric stiffness and stability checks; applied regression tests, numerical tolerances and convergence criteria.

Android / Flutter Developer · Freelancer.com (Remote) · 2017 – present

- Built mobile, desktop and web applications with Java, Kotlin and Dart/Flutter, including APIs, databases, automated tests and deployment workflows.

AI Data Annotation & Automation · Independent / Contract · 2025 – 2026

- Built Python workflows for annotation and evaluation using OCR, LLM integration, browser automation, recovery logic and quality controls.

AI Trainer / Data Quality Specialist · Freelance platforms · 2016 – present

- Prompt writing, response evaluation, fact-checking, STEM rubrics and bilingual (Arabic/English) quality review. Project contributor at AfterQuery (Fenrir, Sliver, Sand projects).

SELECTED SOFTWARE PROJECTS

coroFEM · Nonlinear 2D frame finite element solver · Python · 2026

- Corotational frame solver with Newton-Raphson, displacement control and Crisfield arc-length continuation; verified against analytical references and OpenSeesPy.

ModelMesh · Constraint-satisfaction solver · Python · 2026

- AC-3 / AC-4 and global constraints with MRV, dom/wdeg and LCV heuristics; Hypothesis property-based tests.

Railway Secretariat System · Egyptian National Railways · Flutter & Dart · 2025 – 2026

- Production correspondence system with Clean Architecture, SQLite, OCR/PDF import, authentication hardening, automated tests and Docker deployment.

TECHNICAL SKILLS

Game development: Unreal Engine 5, C++, Gameplay Ability System, StateTree, Enhanced Input, replication, PCG, automation testing, Git LFS.

Physics & numerics: FEM/FEA, stiffness assembly, geometric nonlinearity, Newton-Raphson, arc-length methods, modal analysis, numerical linear algebra.

Algorithms & AI: A*, Dijkstra, constraint solving, search heuristics, state machines, Utility AI.

Languages & tools: C/C++, Python (NumPy, SciPy, pytest, Hypothesis), Dart/Flutter, Java/Kotlin, TypeScript, Git, Linux, Docker, CI workflows.

EDUCATION & TRAINING

B.Sc. Computer Science, Cairo University · 2024

Bachelor of Engineering, Mechanical Engineering, Higher Technology Institute (English program) · 2012

Algorithms, Princeton University (online) · Embedded Linux, Luxoft Egypt (2024 – 2025) · Embedded Linux internship, Orange Digital Center with AMIT (2024) · Embedded Systems, National Telecommunication Institute (2019)

LANGUAGES

Arabic: native · English: advanced (TOEIC Reading & Listening)